FINOPTIX

SUMMER PROJECT '25

Assignment -2 | Intro to Algorithmic Trading

Designing a Robust Bitcoin Trading Strategy

Overview

Develop a trading algorithm for Bitcoin (BTC) that diverges from conventional technical indicators. The goal is to create your own innovative strategies either through purely mathematical logic or by creatively combining existing indicators paired with effective risk management techniques and a back-testing engine.

Data Source & Period:

- Data Provider: Yahoo Finance (yfinance)
- Ticker: BTC-USDT
- Date Range: January 1, 2018 – December 1, 2022
- Initial Capital:₹10,00,000

Trading Strategy Development

- Signal Definitions:
  - Buy Signal: 1
  - Sell Signal: -1
  - No Position: 0
- Core Constraints:
  - Adopt only one active trade at any given time (no overlapping positions).

Risk Management Implementation

Implement at least two of the following (or introduce your own risk management measures):

- Traditional stop loss / take profit levels
- Trailing stop loss
- Dynamic exit conditions
- ATR-based stop loss
- Any other innovative risk management techniques

Graphical Visualization

Develop clear and attractive graphs that visually represent the trading signals on the price chart:

- Buy Signals: Marked with green triangles.
- Sell Signals: Marked with red triangles.

Ensure the graphs are well-labeled, include legends, and are easy to interpret.

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Back-Testing Engine Development

Create a robust back-testing framework to evaluate your trading strategy. This engine should compute and display at least the following performance metrics:

- Benchmark Returns , Return Percentage (%)
- Total Trades Executed
- Win Rate and Losing Trades
- Total Long Trades
- Total Short Trades
- Average Drawdown
- Maximum Drawdown
- Sharpe Ratio
- Sortino Ratio
- Average Holding Time
- Maximum Holding Time

Deliverables

Your final submission should include:

- **Source Code:**
  - Clear, well-documented Python code that performs data extraction (using yfinance), trading strategy computation, risk management application, and back-testing evaluation.

- **Performance Metrics:**
  - A text file displaying all back-test performance metrics.

- **Project Report:**
  - A concise report (in PDF format) explaining your strategy, the rationale behind it, risk management measures used, performance evaluation, and any challenges or insights encountered during development.